CS 254 Machine Learning
Project Proposal Report

1. Introduction

In modern textile manufacturing, fabric quality maintenance is a critical step, but detection of defects is typically a serious bottleneck process. Traditionally, this relies on extensive hand inspection, whereby workers visually inspect cloth for defects, a process both time and prone to human error. This step is thus highly critical to automate, as it has the potential to hugely boost production efficiency and consistency. This project proposes an automated system by way of a machine learning model that detects such production faults directly from images captured by camera systems.

Our overall plan is to design and train a Convolutional Neural Network (CNN) using the TensorFlow Keras Sequential API. This will be a baseline system, chosen for its proven history of application in image classification problems and its ease of implementation by following well-documented TensorFlow tutorials. The model will be trained on the MVTec Anomaly Detection (AD) dataset, which is a set of labeled 32x32 fabric patches that are either "good" or "damaged". In this first proof of concept, we have taken the liberty of simplifying the nature of the problem by aggregating five different types of defects into a single "damaged" class.

This project will demonstrate the feasibility of using a CNN in textile defect detection. The baseline model at the beginning has good accuracy over training data but has issues with the class imbalance of the data set and has a low F1-score on the minority class. Our aim is to dramatically enhance the baseline by using methods like data augmentation and balancing classes to build a stronger and more accurate classifier. Long-term objective is to establish a model not only capable of discriminating good and ruined fabric but expandable to further categorize the specific faults for the final presentation.

2. Problem Definition and Algorithm

2.1 Task Definition

When cloth is produced in factories, a lot of different mishaps might happen. Currently it takes a lot of manual labour to filter out errors in production. Using camera systems to record images of cloth while they are being produced, errors could be detected automatically through a pretrained model. To train the model, the input consists of labeled images of cloth either in a good or a damaged state. To simplify training time, the images will be 64x64 or 32x32 patches.

2.2 Dataset

The data is based on the dataset "The MVTec anomaly detection dataset (MVTec AD)" by the company MVTec and was prepared by Belkhir for Kaggle. It contains images of fabric split into training and test sets with non-overlapping sources. From the original high-definition images, 64x64 patches are generated. Each patch is further rotated at eight fixed angles from 0 to 140 degrees to support rotation-robust classification.

The data is labeled in 6 categories: good (no defect), color, cut, hole, thread and metal contamination.

In the first version, we convert the 5 defect categories into one category "damaged". Therefore our classification is simplified to show a proof of concept. Later all categories will be used for the final presentation.

2.3 Algorithm Definition

The model is a Convolutional Neural Network (CNN) built using Keras Sequential API for image classifications. This CNN serves as a baseline system, and was chosen because of the tensorflow image classification template. The model learns spatial patterns (defects) through convolution and pooling, then classifies using fully connected layers. Future improvements may include class balancing (class_weight or oversampling), data augmentation, or deeper architectures (e.g., transfer learning).

3. Experimental Evaluation

3.1 Methodology

To evaluate our baseline model, we used common metrics like accuracy, precision, recall and F1-score. Because the dataset is pretty imbalanced (about 81% damaged and 19% good), accuracy alone didn't really show the real performance. So we mostly looked at the F1-score for the "good" class since it's the smaller one and harder to detect.

We expected that a simple CNN trained on the MVTec AD dataset would get high accuracy but have problems with the minority class. To test this, we trained it for 10 epochs on 32x32 fabric images, using 80% for training and 20% for validation. We tracked accuracy and loss per epoch and tested it on a separate test set after.

Adn we think that this way makes sense because it directly checks how well the model tells apart damaged and good fabrics, which is the main goal. We also compared our results with similar CNN models in other works that usually reach around 80–90% accuracy, so it's a fair baseline.

3.2 Results

The quantitative results of the baseline model are shown below.

```
F1 (binary): 0.47505270555165147
              precision    recall  f1-score   support

     damaged       0.89      0.91      0.90     30000
        good       0.50      0.45      0.48      6000

    accuracy                           0.83     36000
   macro avg       0.70      0.68      0.69     36000
weighted avg       0.83      0.83      0.83     36000
```

Those are screenshots of the final classification report and training and validation accuracy/loss curves for easier reference.

3.3 Discussion

Our hypothesis was supported because CNN got very high accuracy but had problems recognizing the minority “good” class because of the dataset imbalance. The model kinda overfits, having almost perfect training accuracy but much lower test F1-score. This happens because the model learns dominant damaged patterns easier and mostly ignores the few good samples. By the end of the semester, we plan to improve the

baseline, we may try to use some data augmentation and class balancing to make the model more fair and reliable for real fabric inspection.

4. Related Work

"A Lightweight Approach to Comprehensive Fabric Anomaly Detection Modeling"
In this paper, they address the high computational cost of deep-learning models for fabric defect detection. Their method, GH-YOLOx, uses lightweight modules, pruning, and knowledge distillation to improve speed and reduce parameters while maintaining accuracy. This approach could be worth looking into and comparing against what we're already doing, and we shouldn't immediately assume that our approach will be superior.

"Textile Anomaly Detection: Evaluation of the State-of-the-Art for Automated Quality Inspection of Carpet"
This study evaluates state-of-the-art unsupervised anomaly detection methods for carpet inspection, comparing models like student–teacher and generative approaches. They found student–teacher networks most effective, especially with multi-texture training. Our project could combine student–teacher methods with lightweight architectures for real-time use, and emphasize low false-alarm rates through cost-sensitive optimization.

5. Next Steps

- Marko Tatic: reduce dataset imbalance. Right now, damaged patches make up most of the data. By introducing rotated and slightly changed good images, the data could be balanced (Data augmentation). Also reduce overfitting through different means.
- James Bouchat: Our dataset includes > 15,000 images, around 81% of images have some sort of defect, so our data needs considerable balancing. Right now we're also suffering a bit from overfitting, so experimenting with our regularization methods would be wise.
- Alexander: Explore other model architecture options. The baseline is the current CNN model. I will explore and implement deeper models (e.g., transfer learning with pre-trained ResNet or VGG model) as described in the algorithm

definition. This will be compared with the improved baseline to see if a more complex model can yield higher accuracy and better generalization to the imbalanced dataset.

- Kyrylo Kolesnichenko: I will try different model versions to see which ones work better and compare their results. I also plan to test some changes in the model setup and training to check how it affects accuracy and balance. The goal is to find a solution that performs most stably, efficiently, and generalizes better.

6. Code and Dataset

The jupyter notebook can be found under this GitHub repo:
https://github.com/Chuck953/MachineLearning-UVM

To reproduce, download following dataset and extract the zip file:
https://www.kaggle.com/datasets/belkhirnacim/textiledefectdetection

Then edit "path_to_data" variable to the extracted folder. If the path is correct, the code should run just fine.

7. Conclusion

As a result, we trained a simple CNN on the MVTec AD dataset to detect fabric defects and proved that it can reach high accuracy but still has some problems with the minority "good" class because of imbalance. Overall, now we have a solid baseline and know that CNNs work for this kind of task, but we need to improve the balance and generalization. As was said earlier, in the next phase, we plan to test data augmentation and class weighting to make the model more reliable.

8. Bibliography

Cui, S., Liu, W., & Li, M. (2025). A Lightweight Approach to Comprehensive Fabric Anomaly Detection Modeling. *Sensors*, *25*(7), 2038.
https://doi.org/10.3390/s25072038

Forsberg, B (2024). Textile Anomaly Detection: Evaluation of the State-of-the-Art for Automated Quality Inspection of Carpet. https://www.researchgate.net/publication/382638594_Textile_Anomaly_Detection_Evaluation_of_the_State-of-the-Art_for_Automated_Quality_Inspection_of_Carpet

Danish, M (2022). Textile Defect Detection: Binary Classification. *Kaggle*. https://www.kaggle.com/code/muhmddanish/textile-defect-detection-binary-classification#Textile-Defect-Detection-Project